Answers and talk notes

Determine Distance on the Coordinate Plane

What your student is learning

Tonight we find two distances on a grid and compare them — deciding each time whether to add or subtract the absolute values.

Answers

Ask your student to explain before you look.

1. **A.** The x-coordinate. The x-coordinate is shared: both points have x = 3.
2. **B.** 5. Both y-values are positive, so subtract: 6 – 1 = 5.
3. **D.** 9. The x-values are on opposite sides of 0, so add: 4 + 5 = 9.
4. **A.** 6.5. The y-values are on opposite sides of 0, so add: 4 + 2.5 = 6.5.
5. **B.** Ana, by 2 units. Ana, by 2 units: Ana walks 5 + 3 = 8 units and Beto walks 6 units.

Talk about it

Which coordinate does each trip share? Do you add or subtract for each trip, and who goes farther?

Home mission

Cross two forks on the table to make the axes. Put one coin 3 finger-widths above the crossing and another 5 below it. Ask your student: how far apart are the coins? Add or subtract? Why?

When they are stuck, ask

- Which coordinate do these two points share?
- Do the other two values have the same sign or opposite signs?
- Does the trip cross an axis? Then what do we do?

Try not to say

- Always subtract to find a distance.
- A distance can come out negative, just keep the sign.
- Compare the coordinates instead of the distances.